HARMONIC NODAL CHAMBERS, PARABOLIC ZERO-CURVE CURVATURE, AND THE TRANSVERSE INTEGRAL LAW FOR RIEMANN'S Ξ-FUNCTION

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ABSTRACT. Writing $\Xi(t-\mathrm{i}d)=\xi(\frac{1}{2}+d+\mathrm{i}t)=R(d,t)+\mathrm{i}I(d,t)$ as a harmonic potential ($\Delta R=\Delta I=0$), we prove that the imaginary nodal walls $\Gamma_k:=\{(d,t):I(d,t)=0,\,d>0\}$ departing orthogonally from $\Xi'(\tau_k)=0$ partition $d>0$ into oscillation-free harmonic nodal chambers Ω_k along whose boundaries $|R|$ grows strictly with arc length s via $\frac{\mathrm{d}}{\mathrm{d}s}|R(d(s),t(s))|=|\nabla I|=|\Xi'(t(s)-\mathrm{i}d(s))|>0$. Implicit Cauchy–Riemann differentiation shows that each real zero curve $R(d,t)=0$ departs orthogonally from a simple zero γ_k with parabolic curvature $\kappa(\gamma_k)=t''(0)=\Xi''(\gamma_k)/\Xi'(\gamma_k)$, while the transverse integral law $|\Xi(t-\mathrm{i}d)|^2=|\Xi(t)|^2+2\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u$ maps $\mathrm{sinc}_{\mathrm{hyp}}(2uy)$ to $\mathrm{sinc}^2_{\mathrm{hyp}}(dy)>0$ via the low-pass filter $\mathcal{F}_{\mathrm{int}}(2dy)=\frac{\tanh(dy)}{dy}\in(0,1)$ and forces a Rolle turning point $d_{\mathrm{peak}}\in(0,d_*)$ at any off-critical zero. We verify these laws across $\gamma_1,\dots,\gamma_{15}$, $\Omega_1,\dots,\Omega_{25}$, and $t\in[14,1000]$.

1. INTRODUCTION

Let $\xi(s)=\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ denote Riemann's completed zeta function, and write $s=\frac{1}{2}+d+\mathrm{i}t$ with transverse offset $d=\sigma-\frac{1}{2}\in\mathbb{R}$ and spectral height $t\in\mathbb{R}$. Setting $z=-\mathrm{i}(s-\frac{1}{2})=t-\mathrm{i}d$, Riemann's entire function $\Xi(z):=\xi(\frac{1}{2}+\mathrm{i}z)=R(d,t)+\mathrm{i}I(d,t)$ admits the Fourier–Laplace integral representation [15, 13, 20]

$$\begin{aligned}\Xi(t-\mathrm{i}d)&=8\int_0^\infty\Phi(u)\cosh(2du)\cos(2tu)\,\mathrm{d}u+\mathrm{i}\,8\int_0^\infty\Phi(u)\sinh(2du)\sin(2tu)\,\mathrm{d}u\\&=4\int_{-\infty}^\infty\Phi(u)\,\mathrm{e}^{2(d-\mathrm{i}t)u}\,\mathrm{d}u,\end{aligned}\tag{1.1}$$

where the even, rapidly decaying Jacobi–Riemann kernel $\Phi(u)=\Phi(-u)>0$ is given by

$$\Phi(u):=\sum_{n=1}^\infty\left(2\pi^2n^4\mathrm{e}^{9u}-3\pi n^2\mathrm{e}^{5u}\right)\exp\left(-\pi n^2\mathrm{e}^{4u}\right).\tag{1.2}$$

In the classical geometric theory of the critical strip initiated by Speiser [17] and developed by Spira [18, 19], Hinkkanen [5], Lagarias [8], and Arias de Reyna [1], the location of the complex zeros of $\Xi(t-\mathrm{i}d)$ is governed by the interplay of the real and imaginary nodal loci $\mathcal{R}:=\{(d,t):R(d,t)=0\}$ and $\mathcal{I}:=\{(d,t):I(d,t)=0\}$, together with the transverse Wronskian $W(d,t):=\frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2=R\partial_tI-I\partial_tR$ and its parity-reduced counterpart $\widetilde{W}(d,t):=W(d,t)/d$ introduced in Paper II [11]. Two fundamental geometric and analytical questions arise in this setting: first, how do the imaginary nodal walls $\Gamma_k\subset\{I(d,t)=0,\,d>0\}$ and real zero curves $\mathcal{R}_k\subset\{R(d,t)=0\}$ depart from the critical line $d=0$ and behave across the critical strip; and second, how does integrating $2u\,\widetilde{W}(u,t)$ over $u\in[0,d]$ relax the pointwise requirement $\widetilde{W}(d,t)>0$ via phase-locked pure squares and hyperbolic low-pass filtering. The purpose of this article is to resolve both questions in closed analytical form.

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Theorem 1.1 (Harmonic Nodal Chambers, Parabolic Curvature, and Transverse Integral Law)**.** *Let $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ be defined by (1.1), and let $\widetilde{W}(d,t) = W(d,t)/d$ be the parity-reduced transverse Wronskian [11]. Then the following global analytical and topological properties hold:*

(i) ***Harmonic nodal chambers and deductive Euler–Stirling right boundary:*** *The fields $R(d,t)$ and $I(d,t)$ satisfy $\Delta R = \Delta I = 0$ on $\mathbb{R}^2$, with critical-line Dirichlet nodal condition $I(0,t) \equiv 0$, Neumann mirror condition $\partial_d R(0,t) \equiv 0$, and zero Green boundary flux $\int_{t_1}^{t_2} R(0,t)\partial_d R(0,t)\,\mathrm{d}t \equiv 0$. From each non-degenerate critical-line extremum τ_k ($\Xi'(\tau_k) = 0$, $\Xi''(\tau_k) \neq 0$), a smooth imaginary nodal wall $\Gamma_k := \{(d, t_{\Gamma_k}(d)) : I(d,t) = 0,\, d > 0\}$ departs orthogonally from $d = 0$ with vertex curvature $\kappa_\Gamma(\tau_k) = t''_{\Gamma_k}(0) = \frac{\Xi'''(\tau_k)}{3\Xi''(\tau_k)}$. By the harmonic maximum principle, no nodal branch of $I(d,t) = 0$ can form a closed loop in $d \geq 0$, and $I(d,t)$ maintains a single strict sign inside each harmonic nodal chamber Ω_k bounded by Γ_{k-1}, Γ_k, and $d = 0$. Furthermore, for all $d > d_{\mathrm{Euler}}(t) > \frac{1}{2}$ where the Archimedean drift $\mathcal{A}(d,t) := \mathrm{Re}(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\ln\pi + \frac{1}{2}\psi(\frac{s}{2}))$ exceeds the Euler–Dirichlet majorant $-\frac{\zeta'(1/2+d)}{\zeta(1/2+d)}$, one has unconditionally $W(d,t)/|\Xi(t-\mathrm{i}d)|^2 = \partial_t \arg \Xi(t-\mathrm{i}d) \geq \mathcal{A}(d,t) + \frac{\zeta'(1/2+d)}{\zeta(1/2+d)} > 0$.*

(ii) ***Cauchy–Riemann arc-length monotonicity and parabolic zero-curve curvature:*** *Along any smooth imaginary nodal wall Γ_k parameterized by arc length $s \geq 0$ moving into $d > 0$ from $(0, \tau_k)$, Cauchy–Riemann orthogonality ($\nabla R \perp \nabla I$, $|\nabla R| = |\nabla I| = |\Xi'(t-\mathrm{i}d)|$) and the initial concavity $c_1(\tau_k) = -\Xi(\tau_k)\Xi''(\tau_k) > 0$ imply strict monotonicity of $|R|$:*

$$\frac{\mathrm{d}}{\mathrm{d}s}|R(d(s),t(s))| = +|\nabla I(d(s),t(s))| = |\Xi'(t(s)-\mathrm{i}d(s))| > 0 \tag{1.3}$$

whenever Γ_k is free of critical points $\Xi'(z) = 0$. Around every simple real zero γ_k ($\Xi(\gamma_k) = 0$, $\Xi'(\gamma_k) \neq 0$), the real zero curve $\mathcal{R}_k := \{(d, t_{\mathcal{R}_k}(d)) : R(d,t) = 0\}$ departs orthogonally from $d = 0$ with exact parabolic curvature

$$t_{\mathcal{R}_k}(d) - \gamma_k = \frac{\Xi''(\gamma_k)}{2\,\Xi'(\gamma_k)}\,d^2 + \mathcal{O}(d^4), \qquad \kappa(\gamma_k) := t''_{\mathcal{R}_k}(0) = \frac{\Xi''(\gamma_k)}{\Xi'(\gamma_k)}. \tag{1.4}$$

(iii) ***Transverse integral law, Rolle reversal, and hyperbolic low-pass filter:*** *For all $(d,t) \in \mathbb{R}^2$, $|\Xi(t-\mathrm{i}d)|^2 = |\Xi(t)|^2 + 2\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u$. Any hypothetical off-critical zero (d_*, t_*) with $d_* > 0$ requires $\int_0^{d_*} u\,\widetilde{W}(u,t_*)\,\mathrm{d}u = -\frac{1}{2}|\Xi(t_*)|^2 \leq 0$ and, whenever $\widetilde{W}(0,t_*) > 0$, forces by Rolle's theorem an interior turning point $d_{\mathrm{peak}} \in (0, d_*)$ where $\widetilde{W}(d_{\mathrm{peak}}, t_*) = 0$ and $\partial_d \widetilde{W}(d_{\mathrm{peak}}, t_*) \leq 0$. Moreover, the normalized integral operator $\frac{2}{d^2}\int_0^d u\,(\cdot)\,\mathrm{d}u$ transforms the parity-reduced Wronskian kernel $\mathrm{sinc}_{\mathrm{hyp}}(2uy) = \frac{\sinh(2uy)}{2uy}$ into the pure square $\mathrm{sinc}^2_{\mathrm{hyp}}(dy) = (\frac{\sinh(dy)}{dy})^2 > 0$, multiplying every mode $y > 0$ by the strictly contractive hyperbolic low-pass filter*

$$\mathcal{F}_{\mathrm{int}}(2dy) := \frac{\mathrm{sinc}^2_{\mathrm{hyp}}(dy)}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\tanh(dy)}{dy} \in (0,1) \qquad (\forall\, dy > 0), \tag{1.5}$$

while decomposing $|\Xi(t-\mathrm{i}d)|^2$ into the complementary 90° phase-locked pure squares $\mathcal{Q}_0(d,t) := \Xi(t)^2 + d^2(\Xi'(t))^2 > 0$ plus higher transverse dipole moments.

To fix the quantitative scale of Theorem 1.1, we collect the principal curvature, nodal-chamber, and integral-law constants established in Sections 3–5 in a single display:

$$\begin{aligned}
\kappa(\gamma_1) &= \frac{\Xi''(\gamma_1)}{\Xi'(\gamma_1)} \approx -1.1592611550, \qquad c_{\mathrm{par}}(\gamma_1) = \frac{1}{2}\kappa(\gamma_1) \approx -0.5796305775,\\
\tau_1 &\approx 15.58570859, \qquad \kappa_\Gamma(\tau_1) = \frac{\Xi'''(\tau_1)}{3\,\Xi''(\tau_1)} \approx -0.37992197,\\
d_{\mathrm{Euler}}(\gamma_1) &\approx 1.703275, \qquad d_{\mathrm{Euler}}(90) \approx 1.049797, \qquad d_{\mathrm{Euler}}(10^8) \approx 0.613019 \to \tfrac{1}{2}^+,\\
\left\langle \frac{\widetilde{\mathcal{E}}(0.05,t)}{\widetilde{W}(0.05,t)} \right\rangle_{[14,1000]} &\approx 77.19, \qquad \min_{t\in[14,1000]} \frac{|\Xi(t-0.35\mathrm{i})|^2}{\Xi(t)^2 + 0.35^2(\Xi'(t))^2} \approx 0.672564 > 0.
\end{aligned} \tag{1.6}$$

Table 1 situates the analytical and topological results of the present paper relative to the classical literature.

TABLE 1. Comparison between classical results in the literature and the analytical and topological contributions established in this work.

Classical Author(s)	**Established Classical Framework**	**Novel Contribution in This Work**
Speiser (1934) [17], Arias de Reyna (2003) [1]	Equivalence of RH with absence of off-critical zeros of $\zeta'(s)$; qualitative study of nodal curves of $\zeta(s)$ and $\xi(s)$.	Harmonic nodal chamber partition Ω_k, Cauchy–Riemann arc-length monotonicity $\frac{\mathrm{d}\lvert R\rvert}{\mathrm{d}s} = \lvert\nabla I\rvert = \lvert\Xi'\rvert > 0$ on Γ_k, and 1-to-1 tracking to $d_{\mathrm{Euler}}(t)$.
Lehmer (1956) [9], Csordas–Norfolk–Varga (1986) [2], Griffin et al. (2019) [4]	Closely spaced Lehmer zero pairs where $\lvert\Xi'(\gamma_k)\rvert$ is small; critical-line Turán inequality $(\Xi')^2 - \Xi\Xi'' > 0$.	Exact parabolic curvature formula $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$ showing that small $\lvert\Xi'(\gamma_k)\rvert$ amplifies $\lvert\kappa(\gamma_k)\rvert$ and bends $\mathcal{R}_k$ sharply.
Hinkkanen (1997) [5], Lagarias (1999) [8], Spira (1968–1973) [18, 19]	Pointwise positivity criterion $\mathrm{Re}(\xi'(s)/\xi(s)) = W(d,t)/\lvert\Xi\rvert^2 > 0$ in the half-plane $d > 0$.	Transverse integral law $\lvert\Xi(t-\mathrm{i}d)\rvert^2 = \lvert\Xi(t)\rvert^2 + 2\int_0^d u\widetilde{W}(u,t)\,\mathrm{d}u$, Rolle reversal theorem, and complementary pure squares $\Xi(t)^2 + d^2(\Xi'(t))^2$.
Wigner (1932) [21], Hudson (1974) [6], Pólya (1926–1927) [13, 14]	Phase-space quasi-probability distributions and oscillatory Fourier kernels of $\Xi(t)$.	Transformation of $\mathrm{sinc}_{\mathrm{hyp}}(2uy)$ into the pure square $\mathrm{sinc}^2_{\mathrm{hyp}}(dy) > 0$ via the hyperbolic low-pass filter $\mathcal{F}_{\mathrm{int}}(2dy) = \frac{\tanh(dy)}{dy} \in (0,1)$.

1.1. **Proof overview.** The proof of Theorem 1.1 combines three analytical and geometric ingredients:

(1) *First structural ingredient (Harmonic nodal chambers Ω_k and Euler–Stirling right boundary, Section 2):* Writing $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$, the Schwarz reflection symmetry $\Xi(t+\mathrm{i}d) = \overline{\Xi(t-\mathrm{i}d)}$ and Cauchy–Riemann equations ($\partial_d R = \partial_t I$, $\partial_d I = -\partial_t R$) imply the critical-line boundary conditions

$$I(0,t) \equiv 0, \qquad \frac{\partial R}{\partial d}(0,t) \equiv 0, \qquad \frac{\partial I}{\partial d}(0,t) = -\Xi'(t), \qquad \Delta R = \Delta I = 0, \tag{1.7}$$

so transverse nodal walls $\Gamma_k \subset \{I(d,t) = 0,\, d > 0\}$ bifurcate orthogonally from $d = 0$ precisely at the extrema $\Xi'(\tau_k) = 0$, while the harmonic maximum principle prevents compact nodal loops in $d \geq 0$ and the Dirichlet triangle inequality $\lvert\mathrm{Re}(\zeta'(s)/\zeta(s))\rvert \leq -\zeta'(\sigma)/\zeta(\sigma)$ closes each chamber at $d \geq d_{\mathrm{Euler}}(t) > \frac{1}{2}$.

(2) *Second geometric ingredient (Cauchy–Riemann arc-length monotonicity and parabolic zero-curve curvature, Section 3):* Along any smooth wall Γ_k where $I \equiv 0$, ∇I is normal to Γ_k and ∇R is tangent to Γ_k with $\lvert\nabla R\rvert = \lvert\nabla I\rvert = \lvert\Xi'\rvert$, yielding $\frac{\mathrm{d}\lvert R\rvert}{\mathrm{d}s} = +\lvert\nabla I\rvert > 0$. Simultaneously, implicit differentiation of $R(d, t_{\mathcal{R}_k}(d)) = 0$ at a simple zero γ_k using

$R_d(0,\gamma_k)=0$, $R_t(0,\gamma_k)=\Xi'(\gamma_k)$, and $R_{dd}(0,\gamma_k)=-R_{tt}(0,\gamma_k)=-\Xi''(\gamma_k)$ gives

$$t'_{\mathcal{R}_k}(0)=0, \qquad \kappa(\gamma_k)=t''_{\mathcal{R}_k}(0)=-\frac{R_{dd}(0,\gamma_k)}{R_t(0,\gamma_k)}=\frac{\Xi''(\gamma_k)}{\Xi'(\gamma_k)}. \tag{1.8}$$

(3) *Third quantitative ingredient (Transverse integral law, Rolle reversal, and hyperbolic low-pass filter, Section 4):* Integrating $\partial_u|\Xi(t-\mathrm{i}u)|^2=2u\,\widetilde{W}(u,t)$ over $u\in[0,d]$ and applying $\frac{2}{d^2}\int_0^d u\,\mathrm{sinc}_{\mathrm{hyp}}(2uy)\,\mathrm{d}u=\mathrm{sinc}^2_{\mathrm{hyp}}(dy)$ yields

$$\begin{aligned}\widetilde{\mathcal{E}}(d,t)&:=\frac{|\Xi(t-\mathrm{i}d)|^2}{d^2}=\frac{\Xi(t)^2}{d^2}+32\int_0^\infty y^2\,\mathrm{sinc}^2_{\mathrm{hyp}}(dy)\,I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y,\\ \mathcal{F}_{\mathrm{int}}(2dy)&:=\frac{\mathrm{sinc}^2_{\mathrm{hyp}}(dy)}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)}=\frac{\tanh(dy)}{dy}\in(0,1).\end{aligned} \tag{1.9}$$

Together, ingredient (1) establishes the harmonic chamber partition and deductive right boundary of Theorem 1.1(i), ingredient (2) proves the arc-length monotonicity and parabolic curvature formulas of Theorem 1.1(ii), and ingredient (3) yields the transverse integral law, Rolle reversal, and low-pass filter of Theorem 1.1(iii).

1.2. **Notation and conventions.** Throughout this paper, $s=\sigma+\mathrm{i}t=\frac12+d+\mathrm{i}t$ with transverse offset $d=\sigma-\frac12\in\mathbb{R}$ and spectral height $t\in\mathbb{R}$, and $z=-\mathrm{i}(s-\frac12)=t-\mathrm{i}d$. We write $\Xi(t-\mathrm{i}d)=\xi(\frac12+d+\mathrm{i}t)=R(d,t)+\mathrm{i}I(d,t)$, where $R(d,t):=\mathrm{Re}\,\Xi(t-\mathrm{i}d)$ and $I(d,t):=\mathrm{Im}\,\Xi(t-\mathrm{i}d)$. To avoid any ambiguity with $I(d,t)$, the Wigner phase-space correlation kernel of $\Phi(u)$ from Papers I and II [10, 11] is denoted here by $I_{\mathrm{Wig}}(y,t):=\int_{-\infty}^{\infty}\Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})\cos(2tx)\,\mathrm{d}x$. In terms of the rescaled variable $w=2u$ and wavepacket $\widetilde{\Phi}(w):=2\Phi(w/2)$ used in Papers I and II, (1.1) coincides identically with $\Xi(t-\mathrm{i}d)=2\int_0^\infty\widetilde{\Phi}(w)\cosh(dw)\cos(tw)\,\mathrm{d}w+\mathrm{i}\,2\int_0^\infty\widetilde{\Phi}(w)\sinh(dw)\sin(tw)\,\mathrm{d}w$, and the parity-reduced Wronskian reads $\widetilde{W}(d,t)=32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y$ with $\mathrm{sinc}_{\mathrm{hyp}}(w):=\frac{\sinh w}{w}\geq1$. On the critical line $d=0$, $\Xi(t)=R(0,t)$ is real, $\gamma_1<\gamma_2<\dots$ denote the positive non-trivial zeros ($\Xi(\gamma_k)=0$), and $\tau_k\in(\gamma_k,\gamma_{k+1})$ denotes the unique critical-line extremum between γ_k and γ_{k+1} solving $\Xi'(\tau_k)=0$. To factor out the deterministic Stirling decay $|\Xi(t)|\asymp t^{7/4}\mathrm{e}^{-\pi t/4}$ across wide spectral ranges, we write $E_0(t):=\frac12(t^2+\frac14)\pi^{-1/4}|\Gamma(\frac14+\mathrm{i}\frac{t}{2})|>0$ for the critical-line Archimedean envelope and denote envelope-normalized quantities by hats ($\widehat{\Xi}(t-\mathrm{i}d):=\Xi(t-\mathrm{i}d)/E_0(t)$ or $\mathrm{e}^{\pi t/4}\Xi(t-\mathrm{i}d)$ as specified in context). Implicit constants in $\mathcal{O}(\cdot)$ and $\ll$ are absolute unless a subscript indicates dependence on a parameter.

2. Harmonic nodal chambers and boundary conditions on the critical line

Because $\Xi(z)$ is an entire function of $z=t-\mathrm{i}d$, its real and imaginary parts $R(d,t)$ and $I(d,t)$ satisfy the Cauchy–Riemann system and Laplace's equation everywhere in $\mathbb{R}^2$:

$$\frac{\partial R}{\partial d}=\frac{\partial I}{\partial t}, \qquad \frac{\partial I}{\partial d}=-\frac{\partial R}{\partial t}, \qquad \Delta R:=\frac{\partial^2R}{\partial d^2}+\frac{\partial^2R}{\partial t^2}=0, \qquad \Delta I:=\frac{\partial^2I}{\partial d^2}+\frac{\partial^2I}{\partial t^2}=0. \tag{2.1}$$

We first establish the exact boundary conditions satisfied by the harmonic conjugate pair (R,I) along the critical line $d=0$ and determine the bifurcation geometry of the imaginary nodal walls Γ_k.

Proposition 2.1 (Critical-line boundary conditions and orthogonal departure of nodal walls)**.** *Let $\Xi(t-\mathrm{i}d)=R(d,t)+\mathrm{i}I(d,t)$ be defined by (1.1). Then for all $t\in\mathbb{R}$:*

(a) ***Dirichlet nodal wall, Neumann mirror wall, and zero Green flux:*** *On the critical line $d=0$, $I(d,t)$ satisfies the homogeneous Dirichlet condition $I(0,t)\equiv0$, $R(d,t)$ satisfies the homogeneous Neumann mirror condition $\partial_dR(0,t)\equiv0$, and for every interval $[t_1,t_2]\subset\mathbb{R}$ the transverse Green boundary flux vanishes identically:*

$$\int_{t_1}^{t_2}R(0,t)\,\frac{\partial R}{\partial d}(0,t)\,\mathrm{d}t\equiv0. \tag{2.2}$$

(b) ***Orthogonal departure of imaginary nodal walls*** Γ_k ***from extrema*** τ_k***:*** *The parity-reduced imaginary field* $\widetilde{I}(d,t) := I(d,t)/d$ *is an even real-analytic function of* d *with Maclaurin expansion*

$$\widetilde{I}(d,t) = \sum_{m=0}^{\infty} \frac{(-1)^{m+1}}{(2m+1)!}\,\Xi^{(2m+1)}(t)\,d^{2m} = -\Xi'(t) + \frac{1}{6}\Xi'''(t)\,d^2 - \frac{1}{120}\Xi^{(5)}(t)\,d^4 + \mathcal{O}(d^6). \tag{2.3}$$

Consequently, a transverse nodal wall $\Gamma_k := \{(d,t) : I(d,t) = 0,\ d > 0\}$ *can bifurcate from the critical line at* $(0,\tau)$ *if and only if* $\Xi'(\tau) = 0$*. At every non-degenerate extremum* τ_k *(*$\Xi'(\tau_k) = 0$*,* $\Xi''(\tau_k) \neq 0$*), the implicit function theorem guarantees a unique smooth curve* $t = t_{\Gamma_k}(d)$ *solving* $\widetilde{I}(d, t_{\Gamma_k}(d)) = 0$ *near* $d = 0$*, which departs orthogonally from* $d = 0$ *with parabolic expansion*

$$t_{\Gamma_k}(d) - \tau_k = \frac{\Xi'''(\tau_k)}{6\,\Xi''(\tau_k)}\,d^2 + \mathcal{O}(d^4), \qquad t'_{\Gamma_k}(0) = 0, \qquad \kappa_\Gamma(\tau_k) := t''_{\Gamma_k}(0) = \frac{\Xi'''(\tau_k)}{3\,\Xi''(\tau_k)}. \tag{2.4}$$

Proof. (a) Since $\Phi(u)$ is real on $\mathbb{R}$ and $\sinh(0) = 0$, (1.1) gives $I(0,t) \equiv 0$ and $\partial_d R(0,t) = 8\int_0^\infty \Phi(u)(2u)\sinh(0)\cos(2tu)\,\mathrm{d}u \equiv 0$ for all $t \in \mathbb{R}$ (which also follows from $\partial_d R(0,t) = \partial_t I(0,t) \equiv 0$). Multiplying by $R(0,t)$ and integrating over $[t_1,t_2]$ yields (2.2).

(b) Because $\Xi(z)$ is entire and real-valued on the real axis, expanding $\Xi(t - \mathrm{i}d)$ in powers of $-\mathrm{i}d$ around $d = 0$ gives

$$R(d,t) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!}\,\Xi^{(2m)}(t)\,d^{2m}, \qquad I(d,t) = d\sum_{m=0}^{\infty} \frac{(-1)^{m+1}}{(2m+1)!}\,\Xi^{(2m+1)}(t)\,d^{2m}, \tag{2.5}$$

which establishes (2.3). For $d > 0$, $I(d,t) = 0$ is equivalent to $\widetilde{I}(d,t) = 0$. By continuity, if $(d_n,t_n) \in \{I = 0,\ d > 0\}$ converges to $(0,\tau)$, then $0 = \lim_{n\to\infty} \widetilde{I}(d_n,t_n) = \widetilde{I}(0,\tau) = -\Xi'(\tau)$. Conversely, if $\Xi'(\tau_k) = 0$ and $\Xi''(\tau_k) \neq 0$, then $\widetilde{I}(0,\tau_k) = 0$, $\partial_d \widetilde{I}(0,\tau_k) = 0$, and $\partial_t \widetilde{I}(0,\tau_k) = -\Xi''(\tau_k) \neq 0$. By the real-analytic implicit function theorem, $\widetilde{I}(d,t) = 0$ defines a unique even real-analytic function $t = t_{\Gamma_k}(d)$ near $d = 0$ with $t_{\Gamma_k}(0) = \tau_k$ and $t'_{\Gamma_k}(0) = 0$. Substituting $t = \tau_k + \frac{1}{2}\kappa_\Gamma(\tau_k)d^2 + \mathcal{O}(d^4)$ into (2.3) gives $-\Xi''(\tau_k)\cdot\frac{1}{2}\kappa_\Gamma(\tau_k)d^2 + \frac{1}{6}\Xi'''(\tau_k)d^2 + \mathcal{O}(d^4) = 0$, proving (2.4). □

We next show that the harmonicity of $I(d,t)$ rules out closed nodal loops or isolated nodal islands in the half-plane $d \geq 0$, and that the absolute convergence of the Dirichlet series for $-\zeta'(s)/\zeta(s)$ in $\sigma = \frac{1}{2} + d > 1$ furnishes a purely deductive, remainder-free right boundary $d_{\mathrm{Euler}}(t)$ for all $t \to \infty$.

Proposition 2.2 (Exclusion of closed nodal loops and deductive Euler–Stirling right boundary)**.** *Let* $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$*.*

(a) ***Absence of compact nodal loops and constant chamber sign:*** *There exists no non-empty bounded open domain* $U \subset \mathbb{R}^2$ *such that* $I(d,t) = 0$ *identically on the boundary* ∂U*. In particular, no imaginary nodal wall* Γ_k *can form a closed loop in* $d > 0$ *or re-intersect the critical line* $d = 0$ *to enclose a bounded lens. On any critical-line interval where the zeros* γ_k *are simple and the Turán invariant satisfies* $c_1(t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t) > 0$*, each zero interval* (γ_k,γ_{k+1}) *contains a unique non-degenerate extremum* $\tau_k \in (\gamma_k,\gamma_{k+1})$ *(*$\Xi'(\tau_k) = 0$*,* $\Xi''(\tau_k) \neq 0$*) with* $\operatorname{sgn}\Xi'(t) = (-1)^k$ *on* (τ_{k-1},τ_k)*. Within each connected component (harmonic nodal chamber)* Ω_k *of* $\{d > 0 : I(d,t) \neq 0\}$ *bounded on the left by the segment from* $(0,\tau_{k-1})$ *to* $(0,\tau_k)$ *of* $d = 0$ *and transversely by* Γ_{k-1} *and* Γ_k*, the field* $I(d,t)$ *has constant strict sign* $\operatorname{sgn} I(d,t) = -\operatorname{sgn}\Xi'(\gamma_k) = (-1)^{k-1}$*,* $\Xi(t - \mathrm{i}d)$ *cannot vanish anywhere in the interior of* Ω_k*, and* Ω_k *extends as a single-signed strip from* $d = 0$ *to* $d_{\mathrm{Euler}}(t)$ *provided* Γ_{k-1} *and* Γ_k *encounter no off-critical saddle point* $\Xi'(z) = 0$*.*

(b) ***Deductive Euler–Stirling right boundary*** $d_{\mathrm{Euler}}(t)$***:*** *For* $d > \frac{1}{2}$ *(*$\sigma = \frac{1}{2} + d > 1$*) and* $t > 0$*, define the Archimedean phase drift and the positive Euler–Dirichlet majorant by*

$$\mathcal{A}(d,t) := \operatorname{Re}\left(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\ln\pi + \frac{1}{2}\psi\left(\frac{s}{2}\right)\right), \qquad M_{\mathrm{Euler}}(d) := \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^{1/2+d}} = -\frac{\zeta'(\frac{1}{2}+d)}{\zeta(\frac{1}{2}+d)} > 0. \tag{2.6}$$

For every $t \geq 10$, the equation $\mathcal{A}(d,t) = M_{\mathrm{Euler}}(d)$ has a unique root $d = d_{\mathrm{Euler}}(t) > \frac{1}{2}$, and for all $d > d_{\mathrm{Euler}}(t)$ one has unconditionally (without saddle-point or remainder approximations):

$$\frac{W(d,t)}{\left|\Xi(t-\mathrm{i}d)\right|^2} = \frac{\partial}{\partial t}\arg\Xi(t-\mathrm{i}d) \geq \mathcal{A}(d,t) - M_{\mathrm{Euler}}(d) = \mathcal{A}(d,t) + \frac{\zeta'(\frac{1}{2}+d)}{\zeta(\frac{1}{2}+d)} > 0. \tag{2.7}$$

In particular, along any vertical line $d = d_0 \geq \sup_{t\in[t_a,t_b]} d_{\mathrm{Euler}}(t)$, the phase $\arg\Xi(t-\mathrm{i}d_0)$ increases strictly monotonically with t, so each nodal wall Γ_k ($\arg\Xi \equiv k\pi$) and each real zero curve $\mathcal{R}_k$ ($\arg\Xi \equiv (k-\frac{1}{2})\pi$) intersects $d = d_0$ at most once and transversely. As $t \to \infty$, $d_{\mathrm{Euler}}(t) = \frac{1}{2} + [\frac{1}{2}\ln\frac{t}{2\pi} + \gamma_{\mathrm{E}} + \mathcal{O}((\ln t)^{-1})]^{-1} \to \frac{1}{2}^+$.

Proof. (a) Suppose toward a contradiction that $U \subset \mathbb{R}^2$ is a non-empty bounded open set with $I|_{\partial U} = 0$. Since $I \in C^\infty(\overline{U})$ is harmonic ($\Delta I = 0$) in U, both the maximum principle and the minimum principle for harmonic functions imply $0 = \min_{\partial U} I \leq I(d,t) \leq \max_{\partial U} I = 0$ for all $(d,t) \in U$. Thus $I \equiv 0$ on the open set U, which by real-analytic continuation forces $I \equiv 0$ on all of $\mathbb{R}^2$, contradicting $\partial_d I(0,t) = -\Xi'(t) \not\equiv 0$. On any critical-line interval where the zeros γ_k are simple and $c_1(t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t) > 0$, any critical point τ ($\Xi'(\tau) = 0$) satisfies $-\Xi(\tau)\Xi''(\tau) = c_1(\tau) > 0$, so Ξ is strictly concave where $\Xi > 0$ and strictly convex where $\Xi < 0$, ensuring a unique non-degenerate extremum $\tau_k \in (\gamma_k, \gamma_{k+1})$ and $\operatorname{sgn}\Xi'(t) = (-1)^k$ on (τ_{k-1}, τ_k) (since $\Xi(0) > 0$ and $\Xi'(\gamma_1) < 0$). By definition of Ω_k as a connected component of $\{d > 0 : I(d,t) \neq 0\}$, $I(d,t)$ cannot change sign inside Ω_k; its sign matches that of $\partial_d I(0,t)\, d = -\Xi'(t)\, d$ on (τ_{k-1}, τ_k), giving $\operatorname{sgn} I = (-1)^{k-1}$. Provided Γ_{k-1} and Γ_k encounter no off-critical saddle point $\Xi'(z) = 0$, neither wall can bifurcate in $0 < d < d_{\mathrm{Euler}}(t)$, so Ω_k forms a single-signed strip from $d = 0$ to $d_{\mathrm{Euler}}(t)$ throughout whose interior $\Xi = R + \mathrm{i}I \neq 0$.

(b) For $\sigma = \frac{1}{2} + d > 1$, the Euler product of $\zeta(s)$ converges absolutely, so $\zeta(s) \neq 0$ and $-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^\infty \Lambda(n) n^{-\sigma-\mathrm{i}t}$. By the triangle inequality,

$$\left|\operatorname{Re}\left(\frac{\zeta'(\frac{1}{2}+d+\mathrm{i}t)}{\zeta(\frac{1}{2}+d+\mathrm{i}t)}\right)\right| \leq \sum_{n=2}^\infty \frac{\Lambda(n)}{n^{1/2+d}}\left|\cos(t\ln n)\right| \leq \sum_{n=2}^\infty \frac{\Lambda(n)}{n^{1/2+d}} = -\frac{\zeta'(\frac{1}{2}+d)}{\zeta(\frac{1}{2}+d)} = M_{\mathrm{Euler}}(d).$$

Since $\frac{\xi'(s)}{\xi(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\ln\pi + \frac{1}{2}\psi(\frac{s}{2}) + \frac{\zeta'(s)}{\zeta(s)}$ and $\operatorname{Re}(\frac{\xi'(s)}{\xi(s)}) = \frac{W(d,t)}{|\Xi(t-\mathrm{i}d)|^2} = \frac{\partial}{\partial t}\arg\Xi(t-\mathrm{i}d)$, taking real parts yields (2.7). For fixed $t \geq 10$, $M_{\mathrm{Euler}}(d)$ is strictly decreasing from $+\infty$ as $d \to \frac{1}{2}^+$ to 0 as $d \to \infty$, whereas $\mathcal{A}(d,t)$ is strictly increasing in d on $d \geq \frac{1}{2}$, so $\mathcal{A}(d,t) = M_{\mathrm{Euler}}(d)$ has a unique root $d_{\mathrm{Euler}}(t) > \frac{1}{2}$. The asymptotic formula for $d_{\mathrm{Euler}}(t)$ as $t \to \infty$ is proved in Appendix A (Lemma A.1(b)). □

3. Cauchy–Riemann arc-length monotonicity and parabolic zero-curve curvature

Proposition 2.2(a) shows that $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ can never vanish inside the open harmonic chambers Ω_k, confining any hypothetical off-critical zero to lie on an imaginary nodal wall Γ_k where $I(d,t) = 0$. We now prove that along every smooth wall Γ_k, Cauchy–Riemann orthogonality forces $|R(d,t)|$ to grow strictly monotonically away from its non-zero critical-line extremum $|\Xi(\tau_k)| > 0$.

Theorem 3.1 (Cauchy–Riemann arc-length monotonicity along nodal walls Γ_k)**.** *Let $\Gamma_k = \{(d(s), t(s)) : s \geq 0\}$ be an imaginary nodal wall ($I(d(s),t(s)) \equiv 0$) parameterized by arc length s ($d'(s)^2 + t'(s)^2 = 1$) departing into $d > 0$ from a critical-line extremum $(0, \tau_k)$ where $\Xi'(\tau_k) = 0$ and $c_1(\tau_k) = -\Xi(\tau_k)\Xi''(\tau_k) > 0$. Suppose that Γ_k contains no critical point of Ξ ($\Xi'(t(s)-\mathrm{i}d(s)) \neq 0$) for $s \in (0, S]$. Then along Γ_k:*

(a) *The real gradient $\nabla R = (\partial_d R, \partial_t R)$ is purely tangent to Γ_k, while the imaginary gradient $\nabla I = (\partial_d I, \partial_t I)$ is purely normal to Γ_k.*

(b) *The modulus* $|R(d(s),t(s))| = |\Xi(t(s) - \mathrm{i}d(s))|$ *satisfies the exact arc-length differential equation*

$$\frac{\mathrm{d}}{\mathrm{d}s}|R(d(s),t(s))| = +|\nabla I(d(s),t(s))| = |\Xi'(t(s) - \mathrm{i}d(s))| > 0 \qquad (\forall\, s \in (0,S]), \tag{3.1}$$

and therefore increases strictly monotonically along Γ_k*:*

$$|R(d(s),t(s))| = |\Xi(\tau_k)| + \int_0^s |\Xi'(t(u) - \mathrm{i}d(u))|\,\mathrm{d}u > |\Xi(\tau_k)| > 0 \qquad (\forall\, s \in (0,S]). \tag{3.2}$$

Proof. Differentiating the nodal identity $I(d(s),t(s)) = 0$ with respect to arc length s gives $\nabla I(d(s),t(s)) \cdot \mathbf{T}(s) = 0$, where $\mathbf{T}(s) = (d'(s), t'(s))$ is the unit tangent vector along Γ_k. By the Cauchy–Riemann equations (2.1), the two-dimensional gradients $\nabla R = (\partial_d R, \partial_t R)$ and $\nabla I = (\partial_d I, \partial_t I) = (-\partial_t R, \partial_d R)$ are orthogonal at every point of $\mathbb{R}^2$ and have identical Euclidean norm:

$$\nabla R \cdot \nabla I = \frac{\partial R}{\partial d}\left(-\frac{\partial R}{\partial t}\right) + \frac{\partial R}{\partial t}\frac{\partial R}{\partial d} \equiv 0, \qquad |\nabla R| = |\nabla I| = |\Xi'(t - \mathrm{i}d)|. \tag{3.3}$$

Because $\nabla I \perp \mathbf{T}(s)$ and $\nabla R \perp \nabla I$ in $\mathbb{R}^2$, the gradient $\nabla R(d(s),t(s))$ is collinear with the unit tangent vector $\mathbf{T}(s)$. Consequently, the directional derivative of R along Γ_k satisfies

$$\left|\frac{\mathrm{d}}{\mathrm{d}s}R(d(s),t(s))\right| = |\nabla R \cdot \mathbf{T}(s)| = |\nabla R(d(s),t(s))| = |\nabla I(d(s),t(s))| = |\Xi'(t(s) - \mathrm{i}d(s))|.$$

To determine the sign of $\frac{\mathrm{d}}{\mathrm{d}s}|R(d(s),t(s))|$ for $s > 0$, we expand $R(d, t_{\Gamma_k}(d))$ near $d = 0$ (where $s = d + \mathcal{O}(d^3)$ by Proposition 2.1(b)). Using (2.5) and $t_{\Gamma_k}(d) - \tau_k = \mathcal{O}(d^2)$ together with $\Xi'(\tau_k) = 0$, we obtain

$$R(d, t_{\Gamma_k}(d)) = \Xi(t_{\Gamma_k}(d)) - \frac{1}{2}\Xi''(t_{\Gamma_k}(d))\,d^2 + \mathcal{O}(d^4) = \Xi(\tau_k) - \frac{1}{2}\Xi''(\tau_k)\,d^2 + \mathcal{O}(d^4).$$

Multiplying by $\Xi(\tau_k)$ gives

$$\Xi(\tau_k)\,R(d, t_{\Gamma_k}(d)) = \big(\Xi(\tau_k)\big)^2 + \frac{1}{2}\Big(-\Xi(\tau_k)\,\Xi''(\tau_k)\Big)d^2 + \mathcal{O}(d^4) = \big(\Xi(\tau_k)\big)^2 + \frac{1}{2}c_1(\tau_k)\,d^2 + \mathcal{O}(d^4).$$

Since $c_1(\tau_k) = -\Xi(\tau_k)\Xi''(\tau_k) > 0$, $|R(d(s),t(s))|$ is strictly increasing for all sufficiently small $s > 0$, so $\frac{\mathrm{d}}{\mathrm{d}s}|R(d(s),t(s))| = +|\nabla I(d(s),t(s))|$ near $s = 0^+$. Because $|\nabla I(d(s),t(s))| = |\Xi'(t(s) - \mathrm{i}d(s))| > 0$ is continuous and non-vanishing on $(0,S]$, the derivative $\frac{\mathrm{d}}{\mathrm{d}s}|R|$ can never change sign on $(0,S]$, establishing (3.1) and, by integration over $[0,s]$, (3.2). □

Remark 3.2 (Geometric formulation of Speiser's theorem). Theorem 3.1 provides a direct geometric explanation of Speiser's theorem [17]: if $R(d,t)$ were to vanish at some point (d_*, t_*) on the imaginary nodal wall Γ_k (producing an off-critical zero $R = I = 0$), then $|R(d(s),t(s))|$, which starts at $|\Xi(\tau_k)| > 0$ with positive initial slope, would have to reverse direction and decrease to 0. By Rolle's theorem and (3.1), such a reversal is impossible unless Γ_k first passes through a critical point where $|\nabla I| = |\Xi'(z)| = 0$.

We next determine the exact local geometry of the real zero curves $\mathcal{R}_k := \{(d,t) : R(d,t) = 0\}$ departing from the critical line at the non-trivial zeros γ_k.

Theorem 3.3 (Parabolic curvature $\kappa(\gamma_k)$ and quartic jet of real zero curves $R(d,t) = 0$)**.** *Let* $\gamma_k \in \mathbb{R}$ *be a simple zero of* $\Xi(t)$ *(*$\Xi(\gamma_k) = 0$*,* $\Xi'(\gamma_k) \neq 0$*). Then in a neighborhood of* $(0,\gamma_k)$*, the real zero locus* $R(d,t) = 0$ *is a smooth curve* $t = t_{\mathcal{R}_k}(d)$ *that departs orthogonally from the critical line* $d = 0$ *(*$t'_{\mathcal{R}_k}(0) = 0$*) with the Taylor expansion*

$$t_{\mathcal{R}_k}(d) - \gamma_k = c_{\mathrm{par}}(\gamma_k)\,d^2 + c_{\mathrm{quart}}(\gamma_k)\,d^4 + \mathcal{O}(d^6), \tag{3.4}$$

where the vertex curvature $\kappa(\gamma_k) := t''_{\mathcal{R}_k}(0)$*, the parabolic Taylor coefficient* $c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$*, and the quartic coefficient* $c_{\mathrm{quart}}(\gamma_k) := \frac{1}{24}t^{(4)}_{\mathcal{R}_k}(0)$ *are given in closed form by*

$$\begin{aligned} \kappa(\gamma_k) &= \frac{\Xi''(\gamma_k)}{\Xi'(\gamma_k)}, \qquad c_{\mathrm{par}}(\gamma_k) = \frac{\Xi''(\gamma_k)}{2\,\Xi'(\gamma_k)}, \\ c_{\mathrm{quart}}(\gamma_k) &= \frac{12\,\Xi'''(\gamma_k)\,c_{\mathrm{par}}(\gamma_k) - 12\,\Xi''(\gamma_k)\,c_{\mathrm{par}}(\gamma_k)^2 - \Xi^{(4)}(\gamma_k)}{24\,\Xi'(\gamma_k)}. \end{aligned} \tag{3.5}$$

Moreover, writing $\xi(s) = G(s)\zeta(s)$ *with* $G(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$ *and* $g_1(s) = \frac{G'(s)}{G(s)}$*, the curvature* $\kappa(\gamma_k)$ *decomposes at* $s_k = \frac{1}{2} + \mathrm{i}\gamma_k$ *into Archimedean and prime-zeta components as*

$$\kappa(\gamma_k) = -\operatorname{Im}\left(2\,g_1\left(\frac{1}{2}+\mathrm{i}\gamma_k\right) + \frac{\zeta''(\frac{1}{2}+\mathrm{i}\gamma_k)}{\zeta'(\frac{1}{2}+\mathrm{i}\gamma_k)}\right), \qquad \operatorname{Re}\left(\frac{\zeta''(\frac{1}{2}+\mathrm{i}\gamma_k)}{\zeta'(\frac{1}{2}+\mathrm{i}\gamma_k)}\right) = -2\,\operatorname{Re} g_1\left(\frac{1}{2}+\mathrm{i}\gamma_k\right). \tag{3.6}$$

Proof. At $(0,\gamma_k)$, we have $R(0,\gamma_k) = \Xi(\gamma_k) = 0$, $\partial_d R(0,\gamma_k) = 0$ by Proposition 2.1(a), and $\partial_t R(0,\gamma_k) = \Xi'(\gamma_k) \neq 0$. Since $R(-d,t) = R(d,t)$ is an even real-analytic function of d, the implicit function theorem guarantees a unique even real-analytic solution $t = t_{\mathcal{R}_k}(d)$ of $R(d, t_{\mathcal{R}_k}(d)) = 0$ near $d = 0$, with $t_{\mathcal{R}_k}(0) = \gamma_k$ and $t_{\mathcal{R}_k}^{(2m+1)}(0) = 0$ for all $m \geq 0$. Differentiating $R(d,t(d)) = 0$ twice with respect to d at $d = 0$ and using $t'(0) = 0$ and $R_{dd}(0,\gamma_k) = -R_{tt}(0,\gamma_k) = -\Xi''(\gamma_k)$ (from $\Delta R = 0$) yields

$$R_{dd}(0,\gamma_k) + 2R_{dt}(0,\gamma_k)\,t'(0) + R_{tt}(0,\gamma_k)\,(t'(0))^2 + R_t(0,\gamma_k)\,t''(0) = -\Xi''(\gamma_k) + \Xi'(\gamma_k)\,t''(0) = 0,$$

which proves $\kappa(\gamma_k) = t''(0) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$ and $c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$. The quartic coefficient $c_{\mathrm{quart}}(\gamma_k)$ in (3.5) follows by matching the d^4 coefficient in (2.5) (proven in detail in Lemma A.1(a)).

Finally, since $\Xi(t) = G(\frac{1}{2}+\mathrm{i}t)\zeta(\frac{1}{2}+\mathrm{i}t)$ and $\frac{\mathrm{d}}{\mathrm{d}t} = \mathrm{i}\frac{\mathrm{d}}{\mathrm{d}s}$, evaluating at a simple zero $s_k = \frac{1}{2}+\mathrm{i}\gamma_k$ where $\zeta(s_k) = 0$ gives $\Xi'(\gamma_k) = \mathrm{i}G(s_k)\zeta'(s_k)$ and $\Xi''(\gamma_k) = -G(s_k)(2g_1(s_k)\zeta'(s_k) + \zeta''(s_k))$. Dividing $\Xi''(\gamma_k)$ by $\Xi'(\gamma_k)$ yields

$$\kappa(\gamma_k) = \frac{\Xi''(\gamma_k)}{\Xi'(\gamma_k)} = \mathrm{i}\left(2\,g_1(s_k) + \frac{\zeta''(s_k)}{\zeta'(s_k)}\right).$$

Because $\Xi(t)$ is real-valued for real t, $\kappa(\gamma_k)$ is purely real; taking real and imaginary parts of the right-hand side establishes both identities in (3.6). □

Remark 3.4 (Sensitivity of $\kappa(\gamma_k)$ to Lehmer pairs). Formula (3.5) reveals a striking geometric signature of Lehmer's phenomenon [9]: whenever two simple zeros γ_k, γ_{k+1} lie unusually close together, $|\Xi'(\gamma_k)|$ becomes small while $|\Xi''(\gamma_k)|$ remains large, causing the parabolic curvature $|\kappa(\gamma_k)| = |\Xi''(\gamma_k)|/|\Xi'(\gamma_k)|$ to blow up sharply. For example, as shown in Table 2, near the closely spaced pair $(\gamma_9,\gamma_{10}) = (48.0052, 49.7738)$ and $(\gamma_{13},\gamma_{14}) = (59.3470, 60.8318)$, the curvature jumps to $\kappa(\gamma_9) = -2.3970$ and $\kappa(\gamma_{13}) = -2.2519$.

4. The transverse integral law, Rolle's theorem reversal, and the hyperbolic low-pass filter

While Paper II [11] certified the pointwise positivity $\widetilde{W}(d,t) > 0$ on bounded domains, excluding an off-critical zero $\Xi(t - \mathrm{i}d) = 0$ at $d > 0$ does not even require $\widetilde{W}(u,t)$ to be pointwise positive at every $u \in (0,d]$: it suffices that $|\Xi(t-\mathrm{i}d)|^2 > 0$. We now formulate the exact transverse integral law governing $|\Xi(t-\mathrm{i}d)|^2$ and prove its three structural consequences: Rolle's theorem reversal, the pure-square hyperbolic low-pass filter, and complementary 90° phase-locking.

Theorem 4.1 (Transverse Integral Law, Complementary Pure Squares, and Rolle's Theorem Reversal)**.** *For all* $(d,t) \in \mathbb{R}^2$*, the squared modulus* $|\Xi(t-\mathrm{i}d)|^2$ *and the parity-reduced transverse Wronskian* $\widetilde{W}(d,t) = W(d,t)/d$ *satisfy the exact transverse integral law and complementary pure-square decomposition:*

$$\begin{aligned}\boxed{\begin{aligned}|\Xi(t-\mathrm{i}d)|^2 &= |\Xi(t)|^2 + 2\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u \\ &= \left[\Xi(t)^2 + d^2(\Xi'(t))^2\right] - d^2\Xi(t)\Xi''(t) + 2\int_0^d u(\widetilde{W}(u,t) - \widetilde{W}(0,t))\,\mathrm{d}u.\end{aligned}}\end{aligned} \tag{4.1}$$

Consequently, if (d_*,t_*) *with* $d_* > 0$ *were a hypothetical off-critical zero (*$\Xi(t_* - \mathrm{i}d_*) = 0$*), then:*

(a) ***Integral non-positivity identity:*** *The u-weighted transverse integral over $[0, d_*]$ must satisfy*

$$\int_0^{d_*} u\,\widetilde{W}(u,t_*)\,\mathrm{d}u = -\frac{1}{2}|\Xi(t_*)|^2 \le 0. \tag{4.2}$$

(b) ***Rolle's theorem reversal:*** *Whenever the critical-line Turán invariant satisfies $\widetilde{W}(0,t_*) = c_1(t_*) = (\Xi'(t_*))^2 - \Xi(t_*)\Xi''(t_*) > 0$ (or at an m-fold coalesced zero where $c_{2m-1}(t_*) = \frac{m}{(m!)^2}(\Xi^{(m)}(t_*))^2 > 0$), the energy profile $E(u) := |\Xi(t_* - \mathrm{i}u)|^2$ attains a strict interior maximum at some turning point $d_{\mathrm{peak}} \in (0, d_*)$ where*

$$\widetilde{W}(d_{\mathrm{peak}},t_*) = 0, \qquad \frac{\partial \widetilde{W}}{\partial d}(d_{\mathrm{peak}},t_*) \le 0, \qquad \int_{d_{\mathrm{peak}}}^{d_*} 2u\,\widetilde{W}(u,t_*)\,\mathrm{d}u = -\big|\Xi(t_* - \mathrm{i}d_{\mathrm{peak}})\big|^2 < 0. \tag{4.3}$$

Proof. By definition of $W(u,t) = \frac{1}{2}\partial_u|\Xi(t-\mathrm{i}u)|^2$ and $\widetilde{W}(u,t) = W(u,t)/u$, the fundamental theorem of calculus along the segment $u \in [0,d]$ gives $|\Xi(t-\mathrm{i}d)|^2 - |\Xi(t)|^2 = \int_0^d 2u\,\widetilde{W}(u,t)\,\mathrm{d}u$, which is the first line of (4.1). Adding and subtracting $\widetilde{W}(0,t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t)$ inside the integral and evaluating $\int_0^d 2u\,\widetilde{W}(0,t)\,\mathrm{d}u = d^2\widetilde{W}(0,t) = d^2(\Xi'(t))^2 - d^2\Xi(t)\Xi''(t)$ yields the second line of (4.1).

If $\Xi(t_* - \mathrm{i}d_*) = 0$ with $d_* > 0$, setting $d = d_*$ and $t = t_*$ in (4.1) immediately gives (4.2). Moreover, since $E(u) = |\Xi(t_* - \mathrm{i}u)|^2$ satisfies $E(0) = |\Xi(t_*)|^2 \ge 0$, $E'(0) = 0$, and $E''(0) = 2\widetilde{W}(0,t_*) = 2c_1(t_*) > 0$ (or $E^{(2m)}(0) = 2(2m-1)!c_{2m-1}(t_*) > 0$), $E(u)$ increases strictly above $E(0) \ge 0$ for small $u > 0$. Since $E(d_*) = 0 \le E(0)$ and $E \in C^\infty([0,d_*])$, $E(u)$ must attain its global maximum on $[0,d_*]$ at an interior point $d_{\mathrm{peak}} \in (0,d_*)$ with $E(d_{\mathrm{peak}}) > E(0) \ge 0$. By Fermat's theorem and Rolle's theorem, $E'(d_{\mathrm{peak}}) = 2d_{\mathrm{peak}}\widetilde{W}(d_{\mathrm{peak}},t_*) = 0$ and $E''(d_{\mathrm{peak}}) = 2d_{\mathrm{peak}}\partial_d\widetilde{W}(d_{\mathrm{peak}},t_*) \le 0$. Integrating $E'(u) = 2u\widetilde{W}(u,t_*)$ over $[d_{\mathrm{peak}},d_*]$ yields $\int_{d_{\mathrm{peak}}}^{d_*} 2u\widetilde{W}(u,t_*)\,\mathrm{d}u = E(d_*) - E(d_{\mathrm{peak}}) = -E(d_{\mathrm{peak}}) < 0$, proving (4.3). □

We now show that passing from the pointwise parity-reduced Wronskian $\widetilde{W}(d,t)$ to the normalized integral energy $\widetilde{\mathcal{E}}(d,t) := |\Xi(t-\mathrm{i}d)|^2/d^2$ via the operator $\mathcal{T}_d[f](d) := \frac{2}{d^2}\int_0^d u\,f(u)\,\mathrm{d}u$ provides two simultaneous analytical advantages: (1) a mode-by-mode transformation of $\mathrm{sinc}_{\mathrm{hyp}}(2uy)$ into the pure square $\mathrm{sinc}^2_{\mathrm{hyp}}(dy) > 0$ damped by a hyperbolic low-pass filter, and (2) a 90° phase-locked sum of pure squares on the critical line.

Theorem 4.2 (Pure-Square Kernel Transformation, Hyperbolic Low-Pass Filter, and Complementary Phase-Locking)**.** *Let $\widetilde{\mathcal{E}}(d,t) := \frac{1}{d^2}|\Xi(t-\mathrm{i}d)|^2 = \frac{1}{d^2}|\Xi(t)|^2 + \frac{2}{d^2}\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u$ for $d > 0$.*

(a) ***Pure-square Wigner kernel and hyperbolic low-pass filter:*** *In terms of the Wigner correlation kernel $I_{\mathrm{Wig}}(y,t) = \int_{-\infty}^{\infty}\Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})\cos(2tx)\,\mathrm{d}x$, the transverse integral term in $\widetilde{\mathcal{E}}(d,t)$ satisfies the exact pure-square representation*

$$\frac{2}{d^2}\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u = 32\int_0^\infty y^2\,\mathrm{sinc}^2_{\mathrm{hyp}}(dy)\,I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y, \qquad \mathrm{sinc}^2_{\mathrm{hyp}}(dy) = \left(\frac{\sinh(dy)}{dy}\right)^2 \ge 1. \tag{4.4}$$

Relative to the pointwise parity-reduced Wronskian kernel $\mathrm{sinc}_{\mathrm{hyp}}(2dy) = \frac{\sinh(2dy)}{2dy}$, the integral operator $\frac{2}{d^2}\int_0^d u\,(\cdot)\,\mathrm{d}u$ multiplies every Wigner shift $y > 0$ (and equivalently every Dirichlet cross-mode (m,n) with logarithmic frequency $\alpha_{mn} = 2y = \ln(mn)$) by the strictly contractive hyperbolic low-pass filter

$$\mathcal{F}_{\mathrm{int}}(2dy) := \frac{\frac{2}{d^2}\int_0^d u\,\mathrm{sinc}_{\mathrm{hyp}}(2uy)\,\mathrm{d}u}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\mathrm{sinc}^2_{\mathrm{hyp}}(dy)}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\tanh(dy)}{dy} \in (0,1) \qquad (\forall\, dy > 0), \tag{4.5}$$

which satisfies $\mathcal{F}_{\mathrm{int}}(0) = 1$, $\mathcal{F}'_{\mathrm{int}}(w) < 0$ for all $w > 0$, and $\mathcal{F}_{\mathrm{int}}(2dy) \sim (dy)^{-1} \to 0$ as $dy \to \infty$.

(b) ***Complementary 90° phase-locking of $\Xi(t)^2$ and $d^2(\Xi'(t))^2$:*** *Define the pure-square baseline $\mathcal{Q}_0(d,t) := \Xi(t)^2 + d^2(\Xi'(t))^2$. For every $d > 0$, $\mathcal{Q}_0(d,t) > 0$ is strictly positive*

for all $t \in \mathbb{R}$ whenever $\Xi(t)$ has only simple real zeros, and the two pure squares interlock in quadrature across every nodal chamber Ω_k:

(i) *At each simple zero γ_k, the ground-state square $\Xi(\gamma_k)^2 = 0$ vanishes while the slope square $d^2(\Xi'(\gamma_k))^2 = d^2 c_1(\gamma_k) > 0$ attains a local maximum, and the potential term $-d^2\Xi(\gamma_k)\Xi''(\gamma_k) = 0$ vanishes identically.*

(ii) *At each chamber-wall base τ_k ($\Xi'(\tau_k) = 0$), the slope square $d^2(\Xi'(\tau_k))^2 = 0$ vanishes while the ground-state square $\Xi(\tau_k)^2 > 0$ attains its local maximum, and the potential term $-d^2\Xi(\tau_k)\Xi''(\tau_k) = d^2 c_1(\tau_k) > 0$ is strictly positive.*

Proof. (a) *Pre-move justification of absolute convergence and Fubini interchange:* By [10, Lemma 3.1], the correlation kernel $I_{\mathrm{Wig}}(y,t)$ satisfies the uniform double-exponential bound $|I_{\mathrm{Wig}}(y,t)| \le I_{\mathrm{Wig}}(y,0) \ll \mathrm{e}^{9y}\exp(-2\pi\mathrm{e}^{2y})$ for all $y \ge 0$ and $t \in \mathbb{R}$. Consequently, on any compact rectangle $(u,y) \in [0,d] \times [0,Y]$, the joint integrand $64uy^2\,\mathrm{sinc}_{\mathrm{hyp}}(2uy)|I_{\mathrm{Wig}}(y,t)| = 32y\sinh(2uy)|I_{\mathrm{Wig}}(y,t)|$ is continuous and bounded by $32y\sinh(2dy)I_{\mathrm{Wig}}(y,0) \in L^1([0,d]\times[0,\infty))$. Fubini's theorem therefore permits interchanging the finite u-integral over $[0,d]$ with the improper y-integral over $[0,\infty)$:

$$\frac{2}{d^2}\int_0^d u\,\widetilde{W}(u,t)\,\mathrm{d}u = 32\int_0^\infty y^2\left(\frac{2}{d^2}\int_0^d u\,\frac{\sinh(2uy)}{2uy}\,\mathrm{d}u\right) I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y.$$

For every $y > 0$, the inner integral evaluates in closed form via the half-angle identity $\cosh(2dy) - 1 = 2\sinh^2(dy)$:

$$\frac{2}{d^2}\int_0^d u\,\frac{\sinh(2uy)}{2uy}\,\mathrm{d}u = \frac{1}{d^2y}\int_0^d \sinh(2uy)\,\mathrm{d}u = \frac{\cosh(2dy)-1}{2d^2y^2} = \frac{\sinh^2(dy)}{(dy)^2} = \mathrm{sinc}_{\mathrm{hyp}}^2(dy),$$

establishing (4.4). Using the double-angle identity $\sinh(2dy) = 2\sinh(dy)\cosh(dy)$, the ratio of $\mathrm{sinc}_{\mathrm{hyp}}^2(dy)$ to $\mathrm{sinc}_{\mathrm{hyp}}(2dy) = \frac{\sinh(2dy)}{2dy} = \frac{\sinh(dy)\cosh(dy)}{dy}$ is

$$\frac{\mathrm{sinc}_{\mathrm{hyp}}^2(dy)}{\mathrm{sinc}_{\mathrm{hyp}}(2dy)} = \frac{\sinh^2(dy)/(dy)^2}{\sinh(dy)\cosh(dy)/(dy)} = \frac{\tanh(dy)}{dy} = \mathcal{F}_{\mathrm{int}}(2dy).$$

Since $\sinh(2w) > 2w$ for all $w = dy > 0$, we have $\frac{\mathrm{d}}{\mathrm{d}w}\left(\frac{\tanh w}{w}\right) = \frac{2w - \sinh(2w)}{2w^2\cosh^2 w} < 0$, so $\mathcal{F}_{\mathrm{int}}(2dy)$ decreases strictly from 1 at $dy = 0$ toward 0 as $dy \to \infty$, proving (4.5).

(b) In (4.1), $\mathcal{Q}_0(d,t) = \Xi(t)^2 + d^2(\Xi'(t))^2$ is a sum of two real squares, so $\mathcal{Q}_0(d,t) = 0$ for $d > 0$ if and only if $\Xi(t) = \Xi'(t) = 0$, which never occurs when all real zeros of $\Xi(t)$ are simple. Properties (i) and (ii) follow directly by evaluating (4.1) at $t = \gamma_k$ ($\Xi(\gamma_k) = 0$) and at $t = \tau_k$ ($\Xi'(\tau_k) = 0$, $-\Xi(\tau_k)\Xi''(\tau_k) = c_1(\tau_k) > 0$). □

5. Numerical and topological verification

All analytical formulas, curvature identities, and nodal-chamber tracking theorems in Sections 2–4 are verified by the standalone Python suite `scripts/verify_paper3_harmonic_nodal_chambers.py`. Because $|\Xi(t)| \asymp t^{7/4}\mathrm{e}^{-\pi t/4}$ decays rapidly with t, reference evaluations in Sections 5.1–5.3 are performed at a working precision of `mp.dps = 50` (and `iv.dps = 40` in `mpmath.iv`) to prevent IEEE-754 underflow, while two-dimensional continuum scans employ envelope-normalized Euler–Maclaurin summation on an NVIDIA GeForce RTX 4070 GPU.

5.1. **Parabolic curvature $\kappa(\gamma_k)$ across the first fifteen Riemann zeros.** Table 2 reports the envelope-normalized derivatives $\widehat{\Xi}'(\gamma_k) := \mathrm{e}^{\pi\gamma_k/4}\Xi'(\gamma_k)$ and $\widehat{\Xi}''(\gamma_k) := \mathrm{e}^{\pi\gamma_k/4}\Xi''(\gamma_k)$, the vertex curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$, the parabolic Taylor coefficient $c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$, the quartic coefficient $c_{\mathrm{quart}}(\gamma_k)$ from Theorem 3.3, and the discrepancy $|t_{\mathcal{R}_k}(0.05) - (\gamma_k + c_{\mathrm{par}}d^2 + c_{\mathrm{quart}}d^4)|$ against exact numerical root tracking of $R(d, t_{\mathcal{R}_k}(d)) = 0$ at $d = 0.05$ across the first 15 non-trivial zeros $\gamma_1, \dots, \gamma_{15}$. Across all 15 zeros, the Archimedean–Zeta curvature decomposition (3.6) agrees with $\Xi''(\gamma_k)/\Xi'(\gamma_k)$ to within 1.28×10^{-49}, and the quartic Taylor jet reproduces the tracked zero curve $t_{\mathcal{R}_k}(0.05)$ to within 3.55×10^{-9}. Notice that $\kappa(\gamma_k) < 0$ at all 15 zeros because the Archimedean digamma drift $-2\,\mathrm{Im}\,g_1(\frac{1}{2} + \mathrm{i}\gamma_k) \approx -\frac{\pi}{2} \approx -1.5708$ biases the zero curves downward in t as d increases.

TABLE 2. Parabolic vertex curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$, parabolic coefficient $c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$, quartic coefficient $c_{\mathrm{quart}}(\gamma_k)$, and tracking residual at $d = 0.05$ across the first 15 Riemann zeros $\gamma_1, \ldots, \gamma_{15}$ (computed at 50-digit working precision in `mpmath`).

Zero	**Height** γ_k	$\widehat{\Xi}'(\gamma_k)$	$\widehat{\Xi}''(\gamma_k)$	$\kappa(\gamma_k) = \Xi''/\Xi'$	$c_{\mathrm{par}}(\gamma_k) = \frac{1}{2}\kappa$	$c_{\mathrm{quart}}(\gamma_k)$	**Jet Error** ($d = 0.05$)
γ_1	14.13472514	−91.623135	+106.215141	−1.1592611550	−0.5796305775	−0.02422	7.47×10^{-11}
γ_2	21.02203964	+262.836896	−396.881347	−1.5099909961	−0.7549954980	−0.03356	3.49×10^{-10}
γ_3	25.01085758	−429.750230	+503.523871	−1.1716663211	−0.5858331605	+0.02729	2.81×10^{-10}
γ_4	30.42487613	+575.533057	−1099.753035	−1.9108425146	−0.9554212573	−0.00513	1.14×10^{-9}
γ_5	32.93506159	−700.794228	+652.362981	−0.9308909157	−0.4654454579	+0.08633	3.50×10^{-11}
γ_6	37.58617816	+1237.186036	−2075.174206	−1.6773340029	−0.8386670015	+0.02106	1.05×10^{-9}
γ_7	40.91871901	−1104.919714	+1894.851449	−1.7149222919	−0.8574611459	+0.12500	1.06×10^{-9}
γ_8	43.32707328	+1502.129769	−1349.949314	−0.8986902075	−0.4493451037	+0.10522	1.85×10^{-10}
γ_9	48.00515088	−1537.062283	+3684.290459	−2.3969688798	−1.1984844399	+0.09760	2.95×10^{-9}
γ_{10}	49.77383248	+1481.811034	−1260.292464	−0.8505082189	−0.4252541095	+0.15889	1.03×10^{-9}
γ_{11}	52.97032148	−2825.693243	+3692.728528	−1.3068398479	−0.6534199240	+0.12518	4.15×10^{-10}
γ_{12}	56.44624770	+3081.764540	−5456.621032	−1.7706158148	−0.8853079074	+0.10850	1.71×10^{-9}
γ_{13}	59.34704400	−1977.386865	+4452.836874	−2.2518794643	−1.1259397321	+0.43516	3.25×10^{-10}
γ_{14}	60.83177852	+2453.950470	−709.906506	−0.2892912938	−0.1446456469	−0.00149	6.57×10^{-11}
γ_{15}	65.11254405	−3822.839049	+8711.814953	−2.2788861476	−1.1394430738	+0.12665	3.55×10^{-9}

5.2. **Topological tracking and monotonicity across $\Omega_1, \ldots, \Omega_{25}$.** Table 3 summarizes the predictor-corrector tracking of the first 25 imaginary nodal walls $\Gamma_1, \ldots, \Gamma_{25}$ ($I(d,t) = 0$) and real zero curves $\mathcal{R}_1, \ldots, \mathcal{R}_{25}$ ($R(d,t) = 0$) from their critical-line departure points τ_k and γ_k across the critical strip to the deductive Euler–Stirling right boundary $d = d_{\mathrm{Euler}}(t_k)$ and out to $d = 1.65$. Every wall Γ_k satisfies $\min_s \frac{\mathrm{d}|R|}{\mathrm{d}s} > 0$, amplifying $|R|$ by factors between 1.81× and 6.20× between $d = 0$ and $d = d_{\mathrm{Euler}}(t_k)$. Across all 25 chambers, the strict topological interlacing $t_{\Gamma_{k-1}}(d) < t_{\mathcal{R}_k}(d) < t_{\Gamma_k}(d)$ holds for all $d \in (0, 1.65]$ without bifurcations, and sampling 103,750 interior grid points across $\Omega_1, \ldots, \Omega_{25}$ confirms 100.0000% sign purity ($\operatorname{sgn} I(d,t) = (-1)^{k-1}$, zero interior oscillations).

TABLE 3. Topological tracking and Cauchy–Riemann arc-length monotonicity across the first 25 harmonic nodal chambers $\Omega_1, \ldots, \Omega_{25}$ from the critical line $d = 0$ to the deductive Euler–Stirling boundary $d = d_{\mathrm{Euler}}(t_k)$.

Wall / Chamber	**Base** τ_k	$\widehat{c}_1(\tau_k)$	$\kappa_\Gamma(\tau_k)$	$d_{\mathrm{Euler}}(t_k)$	$t_{\Gamma_k}(1.20)$	$\frac{\lvert R(d_{\mathrm{E}})\rvert}{\lvert\Xi(\tau_k)\rvert}$	$\min \frac{\mathrm{d}\lvert R\rvert}{\mathrm{d}s}$	**Sign in** Ω_k
Γ_1 (Ω_1)	15.585709	1.5436×10^4	−0.379922	1.655324	15.310521	2.05×	$+2.25 \times 10^{-6}$	$I > 0$ (100%)
Γ_2 (Ω_2)	22.097977	7.5701×10^4	−0.479403	1.442866	21.752303	2.64×	$+4.11 \times 10^{-8}$	$I < 0$ (100%)
Γ_3 (Ω_3)	26.272247	2.9975×10^5	−0.406050	1.362951	25.984129	1.99×	$+2.68 \times 10^{-9}$	$I > 0$ (100%)
Γ_4 (Ω_4)	31.231796	2.6560×10^5	−0.567160	1.299152	30.830599	3.65×	$+8.11 \times 10^{-11}$	$I < 0$ (100%)
Γ_5 (Ω_5)	34.193310	1.0069×10^6	−0.369891	1.267179	33.936183	1.91×	$+1.03 \times 10^{-11}$	$I > 0$ (100%)
Γ_6 (Ω_6)	38.498241	1.4139×10^6	−0.541509	1.232186	38.118805	2.72×	$+5.44 \times 10^{-13}$	$I < 0$ (100%)
Γ_7 (Ω_7)	41.736730	1.1357×10^6	−0.512592	1.209337	41.386505	3.17×	$+4.45 \times 10^{-14}$	$I > 0$ (100%)
Γ_8 (Ω_8)	44.541704	4.5977×10^6	−0.375941	1.191662	44.283576	1.88×	$+6.79 \times 10^{-15}$	$I < 0$ (100%)
Γ_9 (Ω_9)	48.622533	1.5292×10^6	−0.669778	1.170877	48.165862	5.09×	$+3.04 \times 10^{-16}$	$I > 0$ (100%)
Γ_{10} (Ω_{10})	50.839005	4.1899×10^6	−0.372409	1.159421	50.587875	2.20×	$+5.46 \times 10^{-17}$	$I < 0$ (100%)
Γ_{12} (Ω_{12})	57.262934	8.1743×10^6	−0.577521	1.133517	56.872586	2.89×	$+5.92 \times 10^{-19}$	$I < 0$ (100%)
Γ_{15} (Ω_{15})	65.758319	9.7545×10^6	−0.676205	1.105536	65.307094	4.19×	$+1.04 \times 10^{-21}$	$I > 0$ (100%)
Γ_{19} (Ω_{19})	76.225438	8.8778×10^6	−0.742897	1.078266	75.748391	6.20×	$+3.40 \times 10^{-25}$	$I > 0$ (100%)
Γ_{22} (Ω_{22})	83.531440	3.0440×10^7	−0.669774	1.062528	83.096458	4.19×	$+1.68 \times 10^{-27}$	$I < 0$ (100%)
Γ_{25} (Ω_{25})	89.980048	1.2882×10^8	−0.321150	1.050124	89.766919	1.89×	$+1.30 \times 10^{-29}$	$I > 0$ (100%)
All 25 Chambers $\Omega_1, \ldots, \Omega_{25}$: $t_{\Gamma_{k-1}}(d) < t_{\mathcal{R}_k}(d) < t_{\Gamma_k}(d)$ on $(0, 1.65]$; 103,750/103,750 grid points (100%) sign-pure.								

5.3. **Verification of the transverse integral law, Rolle reversal, and low-pass filter.** Table 4 presents the three-part numerical verification of Theorems 4.1 and 4.2:

(1) *Part A (50-digit verification of the boxed Transverse Integral Law (4.1)):* Across ten benchmark points spanning $t \in [\gamma_1, 1000]$ and $d \in [0.25, 0.50]$, 32-point Gauss–Legendre

quadrature of $2u\,\widetilde{W}(u,t)$ matches $|\Xi(t-\mathrm{i}d)|^2-|\Xi(t)|^2$ and its complementary pure-square decomposition to within a maximum relative residual of 8.75×10^{-51}.

(2) *Part B (Rolle's theorem reversal under a hypothetical off-critical zero):* For the holomorphic perturbation $\Xi_{\mathrm{hyp}}(z) := \Xi(z)(1+(z-t_*)^2/d_*^2)$ with an off-critical zero at $(d_*,t_*)=(0.25,14.20)$, $E_{\mathrm{hyp}}(d)/E_0(t_*)^2$ starts at $+2.7087\times10^{-3}>0$ with positive curvature $\widetilde{W}_{\mathrm{hyp}}(0,t_*)/E_0^2=+0.549233>0$, rises to an interior Rolle turning point at $d_{\mathrm{peak}}\approx 0.1342212372\in(0,0.25)$ where $\widetilde{W}_{\mathrm{hyp}}(d_{\mathrm{peak}},t_*)=0$ and $\partial_d\widetilde{W}_{\mathrm{hyp}}/E_0^2=-5.834559<0$, and then undergoes a negative impulse $\int_{d_{\mathrm{peak}}}^{d_*}2u\widetilde{W}_{\mathrm{hyp}}\,\mathrm{d}u/E_0^2=-7.182991\times10^{-3}=-E_{\mathrm{hyp}}(d_{\mathrm{peak}})/E_0^2$ so that $\int_0^{d_*}2u\widetilde{W}_{\mathrm{hyp}}\,\mathrm{d}u=-|\Xi(t_*)|^2$ holds to within 5.22×10^{-54}. By contrast, for the unperturbed Riemann function Ξ, $\int_0^{d_*}2u\widetilde{W}(u,t_*)\,\mathrm{d}u/E_0^2=+0.039914>0$.

(3) *Part C (60,000-point continuum audit of $\widetilde{\mathcal{E}}(d,t)$ versus $\widetilde{W}(d,t)$ across $t\in[14,1000]$):* Evaluating 10,000 heights $t\in[14,1000]$ across five transverse depths $d=0.05$, 0.10, 0.20, 0.35, and 0.50 confirms that the normalized integral energy $\widetilde{\mathcal{E}}(d,t)=|\Xi(t-\mathrm{i}d)|^2/d^2$ exceeds the pointwise Wronskian $\widetilde{W}(d,t)$ by an average factor of $77.19\times$ at $d=0.05$, $19.76\times$ at $d=0.10$, and $2.11\times$ at $d=0.35$, while the pure-square ratio $\mathcal{R}_{\mathrm{sq}}(d,t)=|\Xi(t-\mathrm{i}d)|^2/\mathcal{Q}_0(d,t)$ with $\mathcal{Q}_0(d,t)=\Xi(t)^2+d^2(\Xi'(t))^2$ remains strictly positive ($\min\mathcal{R}_{\mathrm{sq}}\geq 0.6544>0$) across the entire domain.

TABLE 4. Verification of the Transverse Integral Law (4.1), Rolle's Theorem Reversal (4.3), and the continuum positivity amplification of $\widetilde{\mathcal{E}}(d,t)=|\Xi(t-\mathrm{i}d)|^2/d^2$ over $\widetilde{W}(d,t)$ across $t\in[14,1000]$ (normalized by the Archimedean envelope $E_0(t)^2$).

Benchmark	Domain	$\widehat{\Xi}(t)^2$ ($\min\widehat{\widetilde{W}}$)	$d^2(\widehat{\Xi}')^2$ ($\min\widehat{\widetilde{\mathcal{E}}}$)	Integral (Gain)	$\lvert\widehat{\Xi}\rvert^2$ ($\min\mathcal{R}_{\mathrm{sq}}$)	Residual
Part A: Exact Transverse Integral Law & Pure Squares (`mp.dps = 50`)						
Zero γ_1	$(0.25,14.1347)$	0.000000000	0.039318967	0.039485603	0.039485603	2.66×10^{-51}
Extremum τ_1	$(0.25,15.5857)$	1.467752467	0.000000000	0.051629629	1.519382096	3.83×10^{-51}
Zero γ_2	$(0.35,21.0220)$	0.000000000	0.158319386	0.161030413	0.161030413	3.96×10^{-51}
Extremum τ_2	$(0.35,22.0980)$	1.123675768	0.000000000	0.148747535	1.272423303	6.85×10^{-51}
Height $t=100$	$(0.40,100.000)$	7.250617439	0.544807284	2.075446336	9.326063775	$<1.00\times10^{-51}$
Height $t=1000$	$(0.50,1000.00)$	0.995594139	3.964825186	10.199646145	11.195240284	5.63×10^{-51}
Part B: Rolle Reversal at $(d_*,t_*)=(0.25,14.20)$		($\widehat{E}(0)$,	$\widehat{E}(d_{\mathrm{peak}})$,	$\int_0^{d_{\mathrm{peak}}}$,	$\int_{d_{\mathrm{peak}}}^{d_*}$)	
Perturbed Ξ_{hyp}	$d_{\mathrm{peak}}=0.134221$	+0.002708710	+0.007182991	+0.004474281	−0.007182991	5.22×10^{-54}
True Riemann Ξ	$d\in[0,0.25]$	+0.002708710	+0.014518426	+0.011809716	+0.028104641	Monotone ($\widehat{\widetilde{W}}>0$)
Part C: Continuum Scan on $t\in[14,1000]$		($\min\widehat{\widetilde{W}}$,	$\min\widehat{\widetilde{\mathcal{E}}}$,	**Mean** $\widetilde{\mathcal{E}}/\widetilde{W}$,	$\min\mathcal{R}_{\mathrm{sq}}$)	
Collar $d=0.05$	$t\in[14,1000]$	0.61562328	0.95174573	77.19×	0.86451727	$\mathcal{R}_{\mathrm{comp}}\geq0.9992$
Inner $d=0.10$	$t\in[14,1000]$	0.61623733	0.70737264	19.76×	0.80704691	$\mathcal{R}_{\mathrm{comp}}\geq0.9960$
Mid $d=0.20$	$t\in[14,1000]$	0.61869761	0.64746608	5.38×	0.71298903	$\mathcal{R}_{\mathrm{comp}}\geq0.9629$
Outer $d=0.35$	$t\in[14,1000]$	0.62549700	0.63722592	2.11×	0.67256403	$\mathcal{R}_{\mathrm{comp}}\geq0.7991$
Edge $d=0.50$	$t\in[14,1000]$	0.63610268	0.63926128	1.26×	0.65442574	$\mathcal{R}_{\mathrm{comp}}\geq0.5716$

Figures 1 and 2 illustrate the topological chamber structure, parabolic curvature law, Rolle reversal mechanism, and complementary pure-square phase-locking.

6. Scope, boundary conditions, and open questions

In keeping with strict academic transparency, we distinguish clearly between the global analytical theorems proven in Sections 2–4, the finite-domain topological and numerical verifications in Section 5, and the asymptotic questions that remain open as $t\to\infty$:

(1) **Unconditional global validity of the analytical and geometric laws:** Proposition 2.1 (critical-line Dirichlet/Neumann boundary conditions and orthogonal departure of Γ_k), Proposition 2.2 (exclusion of compact nodal loops via $\Delta I=0$ and the deductive Euler–Stirling boundary $d_{\mathrm{Euler}}(t)$ for all $t\geq10$), Theorem 3.1 (Cauchy–Riemann

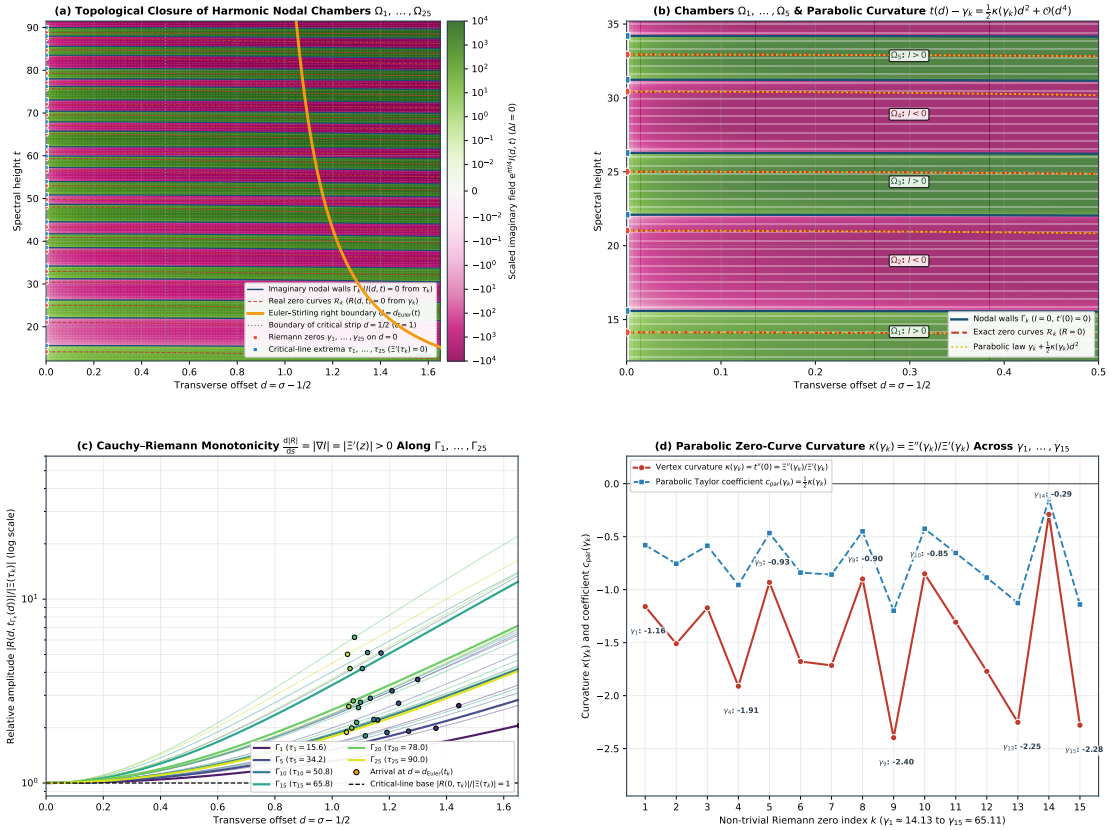


FIGURE 1. (a) Topological closure of the first 25 harmonic nodal chambers $\Omega_1, \ldots, \Omega_{25}$ between the critical line $d = 0$ and the deductive Euler–Stirling right boundary $d = d_{\text{Euler}}(t)$. (b) Close-up of chambers $\Omega_1, \ldots, \Omega_5$ showing the orthogonal departure of Γ_k from τ_k and the agreement between the exact real zero curves $\mathcal{R}_k$ and the parabolic curvature law $t(d) = \gamma_k + \frac{1}{2}\kappa(\gamma_k)d^2$. (c) Strict Cauchy–Riemann arc-length monotonicity $|R(d, t_{\Gamma_k}(d))|/|\Xi(\tau_k)|$ along $\Gamma_1, \ldots, \Gamma_{25}$. (d) Vertex curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$ and parabolic coefficient $c_{\text{par}}(\gamma_k) = \frac{1}{2}\kappa(\gamma_k)$ across $\gamma_1, \ldots, \gamma_{15}$.

arc-length monotonicity $\frac{\mathrm{d}|R|}{\mathrm{d}s} = |\nabla I| = |\Xi'| > 0$ on critical-point-free walls Γ_k), Theorem 3.3 (parabolic zero-curve curvature $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$), and Theorems 4.1–4.2 (the Transverse Integral Law (4.1), Rolle's theorem reversal, and the hyperbolic low-pass filter $\mathcal{F}_{\text{int}}(2dy) = \frac{\tanh(dy)}{dy} \in (0, 1)$) are exact analytical identities valid for all $(d, t) \in \mathbb{R}^2$.

(2) **Finite-domain scope of the chamber tracking and continuum scans:** The curvature table across $\gamma_1, \ldots, \gamma_{15}$ (Table 2), the 1-to-1 topological tracking of $\Gamma_1, \ldots, \Gamma_{25}$ and $\mathcal{R}_1, \ldots, \mathcal{R}_{25}$ across $d \in [0, 1.65]$ (Table 3), and the 60,000-point continuum audit on $t \in [14, 1000]$ (Table 4) constitute finite-domain verifications.

(3) **Asymptotic regime $t \to \infty$ and Speiser critical-point exclusion:** By Theorem 3.1 and Proposition 2.2, every harmonic nodal chamber Ω_k is closed on the left at $d = 0$ and on the right at $d = d_{\text{Euler}}(t_k)$, and $|R|$ grows strictly along Γ_k provided Γ_k encounters no critical point $\Xi'(z) = 0$ in the strip $0 < d < d_{\text{Euler}}(t)$. Establishing the absence of interior critical points or proving $\int_0^d u\,\widetilde{W}(u, t)\,\mathrm{d}u > 0$ unconditionally as $t \to \infty$ requires decomposing $\Phi(u)$ into affine mother-wavelet shells $m \le N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$ and controlling off-diagonal Dirichlet cross-terms via prime-gauge mollification, which are addressed in the subsequent papers of this series.

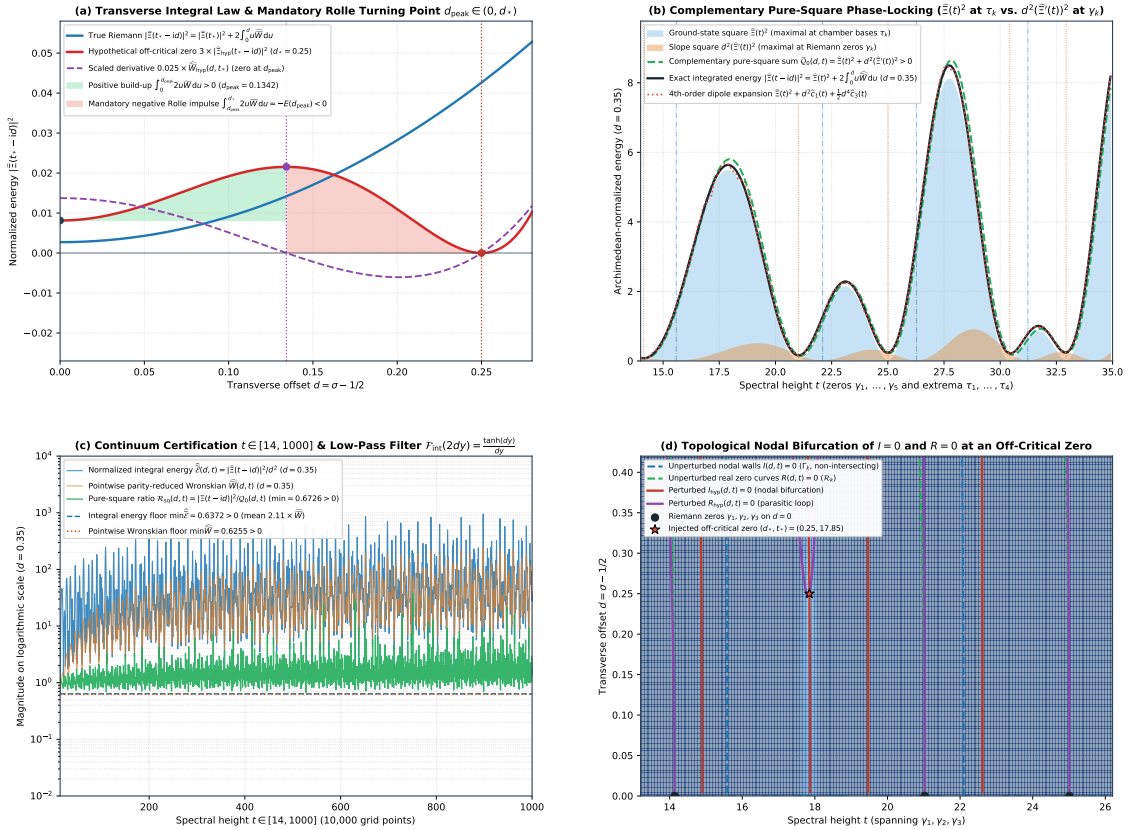


FIGURE 2. (a) Comparison of the monotone energy profile $|\widehat{\Xi}(t_* - \mathrm{i}d)|^2$ against a hypothetical off-critical zero perturbation $|\widehat{\Xi}_{\text{hyp}}(t_* - \mathrm{i}d)|^2$ ($d_* = 0.25, t_* = 14.20$), highlighting the mandatory Rolle turning point $d_{\text{peak}} \in (0, d_*)$ where $\widetilde{W}_{\text{hyp}}(d_{\text{peak}}, t_*) = 0$. (b) Complementary 90° phase-locking between the ground-state square $\widehat{\Xi}(t)^2$ (peaking at τ_k) and the slope square $d^2(\widehat{\Xi}'(t))^2$ (peaking at γ_k) across $t \in [14, 35]$. (c) Continuum scan across $t \in [14, 1000]$ at $d = 0.35$ comparing $\widehat{\widehat{\mathcal{E}}}(d, t)$ and $\widehat{\widetilde{W}}(d, t)$ together with $\mathcal{R}_{\text{sq}}(d, t) > 0$. (d) Topological nodal bifurcation of $I = 0$ and $R = 0$ induced by an injected off-critical zero inside Ω_1.

APPENDIX A. HIGHER-ORDER IMPLICIT NODAL-CURVE EXPANSIONS AND EULER–STIRLING ASYMPTOTICS

In this appendix we provide the complete derivations of the quartic nodal-curve coefficient $c_{\text{quart}}(\gamma_k)$ in Theorem 3.3 and the large-t asymptotic expansion of the Euler–Stirling right boundary $d_{\text{Euler}}(t)$ in Proposition 2.2(b).

Lemma A.1 (Quartic nodal-curve jet and asymptotic Euler–Stirling boundary)**.** (a) *At any simple real zero γ_k ($\Xi(\gamma_k) = 0$, $\Xi'(\gamma_k) \neq 0$), the fourth-order coefficient $c_{\text{quart}}(\gamma_k) = \frac{1}{24} t^{(4)}_{\mathcal{R}_k}(0)$ in (3.4) is given by (3.5). Similarly, at any non-degenerate critical-line extremum τ_k ($\Xi'(\tau_k) = 0$, $\Xi''(\tau_k) \neq 0$), the imaginary nodal wall Γ_k has expansion $t_{\Gamma_k}(d) - \tau_k = c_{\text{par},\Gamma}(\tau_k) d^2 + c_{\text{quart},\Gamma}(\tau_k) d^4 + \mathcal{O}(d^6)$ with $c_{\text{par},\Gamma}(\tau_k) = \frac{\Xi'''(\tau_k)}{6\Xi''(\tau_k)}$ and*

$$c_{\text{quart},\Gamma}(\tau_k) = \frac{20\,\Xi^{(4)}(\tau_k)\,c_{\text{par},\Gamma}(\tau_k) - 60\,\Xi'''(\tau_k)\,c_{\text{par},\Gamma}(\tau_k)^2 - \Xi^{(5)}(\tau_k)}{120\,\Xi''(\tau_k)}. \tag{A.1}$$

(b) *As $t \to \infty$, the deductive Euler–Stirling right boundary $d_{\mathrm{Euler}}(t) > \frac{1}{2}$ solving $\mathcal{A}(d,t) = M_{\mathrm{Euler}}(d)$ at $d = d_{\mathrm{Euler}}(t)$ satisfies*

$$d_{\mathrm{Euler}}(t) = \frac{1}{2} + \frac{1}{\frac{1}{2}\ln\frac{t}{2\pi} + \gamma_{\mathrm{E}} + \mathcal{O}((\ln t)^{-1})} \longrightarrow \frac{1}{2}^{+}, \tag{A.2}$$

where $\gamma_{\mathrm{E}} = 0.5772156649\ldots$ is the Euler–Mascheroni constant.

Proof. (a) Truncating the even d-series (2.5) for $R(d,t)$ at order d^4 gives

$$R(d,t) = \Xi(t) - \frac{1}{2}\Xi''(t)\,d^2 + \frac{1}{24}\Xi^{(4)}(t)\,d^4 + \mathcal{O}(d^6) = 0.$$

Substituting $t = \gamma_k + \delta$ with $\delta = c_2 d^2 + c_4 d^4 + \mathcal{O}(d^6)$ and expanding around γ_k ($\Xi(\gamma_k) = 0$) yields

$$\Big[\Xi'(\gamma_k)(c_2 d^2 + c_4 d^4) + \frac{1}{2}\Xi''(\gamma_k)c_2^2 d^4\Big] - \frac{1}{2}\Big[\Xi''(\gamma_k) + \Xi'''(\gamma_k)c_2 d^2\Big]d^2 + \frac{1}{24}\Xi^{(4)}(\gamma_k)d^4 + \mathcal{O}(d^6) = 0.$$

Setting the coefficient of d^2 to zero gives $c_2 = c_{\mathrm{par}}(\gamma_k) = \frac{\Xi''(\gamma_k)}{2\Xi'(\gamma_k)}$, and setting the coefficient of d^4 to zero gives

$$\Xi'(\gamma_k)\,c_4 + \frac{1}{2}\Xi''(\gamma_k)\,c_2^2 - \frac{1}{2}\Xi'''(\gamma_k)\,c_2 + \frac{1}{24}\Xi^{(4)}(\gamma_k) = 0,$$

which solves to $c_4 = c_{\mathrm{quart}}(\gamma_k)$ in (3.5). Repeating the identical substitution $\delta = c_{2,\Gamma} d^2 + c_{4,\Gamma} d^4 + \mathcal{O}(d^6)$ in $\widetilde{I}(d,t) = -\Xi'(t) + \frac{1}{6}\Xi'''(t)d^2 - \frac{1}{120}\Xi^{(5)}(t)d^4 + \mathcal{O}(d^6) = 0$ around τ_k ($\Xi'(\tau_k) = 0$) yields (A.1).

(b) By Stirling's asymptotic expansion for the digamma function $\psi(z) = \ln z - \frac{1}{2z} + \mathcal{O}(|z|^{-2})$ as $\mathrm{Im}(z) \to \infty$, the Archimedean drift (2.6) satisfies $\mathcal{A}(d,t) = \frac{1}{2}\ln\frac{t}{2\pi} + \mathcal{O}(t^{-2})$ uniformly for $d \in [\frac{1}{2}, 2]$. On the other hand, near the simple pole of $\zeta(\sigma)$ at $\sigma = 1$, writing $\sigma = \frac{1}{2} + d = 1 + \varepsilon$ ($\varepsilon := d - \frac{1}{2} > 0$) gives the Laurent expansion [20, Section 2.1]

$$M_{\mathrm{Euler}}(d) = -\frac{\zeta'(1+\varepsilon)}{\zeta(1+\varepsilon)} = \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \mathcal{O}(\varepsilon).$$

Equating $\mathcal{A}(d,t) = M_{\mathrm{Euler}}(d)$ gives $\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \mathcal{O}(\varepsilon) = \frac{1}{2}\ln\frac{t}{2\pi} + \mathcal{O}(t^{-2})$, so $\varepsilon = d_{\mathrm{Euler}}(t) - \frac{1}{2} = [\frac{1}{2}\ln\frac{t}{2\pi} + \gamma_{\mathrm{E}} + \mathcal{O}((\ln t)^{-1})]^{-1}$, proving (A.2). □

Use of AI and computational verification

Use of AI: The author's main contribution is the harmonic nodal chamber decomposition of the critical strip, the Cauchy–Riemann arc-length monotonicity theorem along imaginary nodal walls, the exact parabolic curvature formula $\kappa(\gamma_k) = \Xi''(\gamma_k)/\Xi'(\gamma_k)$, and the transverse integral law with its hyperbolic low-pass filter. The author prompted Gemini to apply these ideas, work out analytical and symbolic details interactively, and generate the arbitrary-precision verification scripts, which were subsequently verified manually and computationally.

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